

VENKATA SAMYUKTA MALAPAKA

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EDUCATION

University of Massachusetts, Amherst

Feb'23 – Apr'25

Master of Science in Computer Science (Data Science Honors)

Jawaharlal Nehru Technological University, Hyderabad, India

Aug'16 – May'20

Bachelor of Technology in Computer Science and Engineering

TECHNICAL SKILLS

AI & LLM Engineering: RAG, Agent Orchestration, Cursor, Prompt Eng, Fine-tuning, LangGraph, Claude Code, Codex

Programming Languages: Python, TypeScript, Java, Scala, JavaScript, SQL, Bash

Backend & Frameworks: FastAPI, Spring Boot, gRPC, REST APIs, Node.js, Next.js, React

Data & Infra: PostgreSQL /pgvector, Redis, Kafka, Spark, Elasticsearch, AWS, Docker, Kubernetes, Terraform, CI/CD

EXPERIENCE

Pipelines - Founding Engineer

Aug 2025 – Present

- Launched the evaluation layer for AI agents: a simulation platform that runs agents through stateful, reactive scenarios by injecting realistic failures for error recovery before production
- Hardened the metric-evaluation algorithms behind an LLM-as-judge verdict engine : rubric-based scoring with bias and consistency controls improving grading reliability and inter-run agreement by 75%
- Owned the SDK end-to-end, integrating any agent in a few lines of code without changes to prompts or models, cutting eval iteration cycles by 80%

Amazon - Software Development Engineer

Apr 2025 – Apr 2026

- Designed and delivered secure, low-latency distributed services for Amazon Payments, strengthening DDoS resilience and ensuring PCI-DSS compliance
- Built asynchronous payment transmission pipelines to issuing banks, reducing end-to-end latency from 15 ms to 7 ms through non-blocking I/O, batching, and service refactoring
- Architected inbound bank-to-Amazon ingestion services end-to-end infrastructure by defining protocols, implementing network security, launched across 3 regions : USA, Spain and India.

Otom AI - Founding Engineer (Co-op)

Jan 2025– June 2025

- Architected a prompt optimization framework using semantic embeddings, context compression, and distributed caching across 6 production chatbot applications
- Improved response accuracy to 95% while reducing inference cost by 40% through hybrid retrieval-augmented generation (RAG) techniques

IgnitelQ - AI Engineer (Co-op)

Oct 2023 – Dec 2023

- Architected a prompt optimization framework using semantic embeddings, context compression, and distributed caching across 6 production chatbot applications
- Improved response accuracy to 95% while reducing inference cost by 40% through hybrid retrieval-augmented generation (RAG) techniques

Verizon - Full Stack Software Development Engineer 2

Feb 2020 – Jan 2023

- Developed Java and Spring Boot microservices for a global 5G field technician scheduling platform, delivering \$5M in measurable business impact
- Improved system reliability by eliminating 2 hours/week of downtime through automated data refresh and recovery pipelines
- Built distributed analytics pipelines using Spark, Hive on HDFS, and the ELK stack to process 18TB+ of production data
- Optimized SQL query performance through indexing, partitioning, and caching, reducing report latency by 90%

PROJECTS

Graph RAG for BioNLP: Social Determinants of Health Analysis

Dec 2024 – Jul 2025

- Built a graph-based RAG pipeline over MIMIC-III/IV (50K+ patient records) using SciSpacy entity linking and BioBERT embeddings, extracting 1M+ clinical entities and relationships across structured & unstructured data
- Improved correlation discovery for social determinants of health by 35% and reduced irrelevant retrievals by 28% via hybrid graph + vector search, enabling interpretable patient risk analysis

YOLOv11-Based Crowd Flow Detection on Image Data

Sep 2024 – Dec 2024

- Designed a bandwidth-efficient crowd flow system using static images, replacing video pipelines with a two-stage YOLOv11 detection & flow classification architecture
- Achieved 90.1% mAP50 and 85% precision in detection, validating live deployment in constrained compute environment

Personalization in Large Language Models

Sep 2023 – Dec 2023

- Fine-tuned the Flan-T5-base architecture using engineered personalized prompts with AdamW optimization, warmup scheduling, linear decay, and L2 regularization over 10 epochs. Strategised using context-aware prompts using BM25-based retrieval, achieving 80% accuracy on personalization benchmarks